Final_US_Public_Transit_Numbers

October 25, 2026

Looking at the top 10 major transit agencies by ridership in the United States when Vehicle Revenue Miles (VRM) is adjusted how does passenger ridership (UPT) respond, and how does this differ between Heavy Rail (subways) and Light Rail Vehicles (LRV's)?

[1]:
```
import pandas as pd
```

[2]:
```
import numpy as np
```

[3]:
```
import matplotlib.pyplot as plt
```

[4]:
```
import seaborn as sns
```

[5]:
```
#Load the csv file and check information
```

[6]:
```
df=pd.read_csv('Downloads/
↪Complete_Monthly_Ridership_(with_Adjustments_and_Estimates)_20261025.csv')
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_1544\2253705021.py:1: DtypeWarning:
Columns (0: Legacy NTD ID, 1: VOMS) have mixed types. Specify dtype option on
import or set low_memory=False.
  df=pd.read_csv('Downloads/Complete_Monthly_Ridership_(with_Adjustments_and_Est
imates)_20261025.csv')
```

[7]:
```
df.head()
```

[7]:

	NTD ID	Legacy NTD ID	Agency	Mode/Type of Service	Status	Reporter Type	UACE CD	UZA Name	Mode	TOS	3 Mode	Date
0	1	0001	King County		Active	Full Reporter	80389.0	Seattle--Tacoma, WA	DR	PT	Bus	January 2002
1	1	0001	King County		Active	Full Reporter	80389.0	Seattle--Tacoma, WA	DR	PT	Bus	February 2002
2	1	0001	King County		Active	Full Reporter	80389.0	Seattle--Tacoma, WA	DR	PT	Bus	March 2002
3	1	0001	King County		Active	Full Reporter	80389.0	Seattle--Tacoma, WA	DR	PT	Bus	April 2002
4	1	0001	King County		Active	Full Reporter	80389.0	Seattle--Tacoma, WA	DR	PT	Bus	May 2002

```
       UPT VOMS     VRH      VRM  \
0  135,144  574  53,306  746,158
1  127,378  574  46,865  656,324
2  136,030  574  51,889  726,578
3  142,204  574  52,674  736,975
4  144,697  574  53,306  746,158

                                Agency Mode TOS Date State  FTA Region
0  00001_DR_PT_Active_Full Reporter_80389_2002-01-01    WA           10
1  00001_DR_PT_Active_Full Reporter_80389_2002-02-01    WA           10
2  00001_DR_PT_Active_Full Reporter_80389_2002-03-01    WA           10
3  00001_DR_PT_Active_Full Reporter_80389_2002-04-01    WA           10
4  00001_DR_PT_Active_Full Reporter_80389_2002-05-01    WA           10
```

[8]:
```
df.shape
```

[8]:
```
(372801, 18)
```

[9]:
```
df.describe()
```

[9]:
```
              NTD ID        UACE CD     FTA Region
count  372801.000000  372741.000000  372801.000000
mean    47879.018020   50031.179546       5.497102
std     27389.850199   28637.056371       2.698013
min         1.000000      17.000000       1.000000
25%     30027.000000   22960.000000       4.000000
50%     50008.000000   51755.000000       5.000000
75%     70003.000000   75718.000000       9.000000
max     99479.000000   99999.000000      10.000000
```

[10]:
```
# Verification of data before transformation
```

[11]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 372801 entries, 0 to 372800
Data columns (total 18 columns):
 #   Column                        Non-Null Count   Dtype
---  ------                        --------------   -----
 0   NTD ID                        372801 non-null  int64
 1   Legacy NTD ID                 366204 non-null  object
 2   Agency                        372801 non-null  str
 3   Mode/Type of Service Status   372801 non-null  str
 4   Reporter Type                 372801 non-null  str
 5   UACE CD                       372741 non-null  float64
 6   UZA Name                      372741 non-null  str
 7   Mode                          372801 non-null  str
 8   TOS                           372801 non-null  str
```

```
 9   3 Mode                          372801 non-null  str
 10  Date                            372801 non-null  str
 11  UPT                             372801 non-null  str
 12  VOMS                            371086 non-null  object
 13  VRH                             369288 non-null  str
 14  VRM                             369519 non-null  str
 15  Agency Mode TOS Date            372801 non-null  str
 16  State                           372801 non-null  str
 17  FTA Region                      372801 non-null  int64
dtypes: float64(1), int64(2), object(2), str(13)
memory usage: 107.4+ MB
```

[12]:
```
# Data cleanup to cocentrate on the top 10 agencies based on ridership
```

[13]:
```
# Conversion of Ridership (UPT) and Vehicle Revenue Miles (VRM) to integrals␣
 ↪because the data in the csv had them stored as a str for easy display, but␣
 ↪need them to be a value in order to do calculations with.  Same reason for␣
 ↪the date, as it is stored as a str for dispay but will not be easy to use␣
 ↪for caluations in charting. To keep data integrity, orignal values can be␣
 ↪referenced to since integral forms of the VRM and UPT now have their own␣
 ↪columns that I can perform calculations on.
```

[14]:
```
df['UPT_num'] = pd.to_numeric(df['UPT'].astype(str).str.replace(',', ''))
df['VRM_num'] = pd.to_numeric(df['VRM'].astype(str).str.replace(',', ''))
```

[15]:
```
df['Date'] = pd.to_datetime(df['Date'])
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_1544\2394721818.py:1: UserWarning:
Could not infer format, so each element will be parsed individually, falling
back to `dateutil`. To ensure parsing is consistent and as-expected, please
specify a format.
  df['Date'] = pd.to_datetime(df['Date'])
```

[16]:
```
# Verification of data after transformation
```

[17]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 372801 entries, 0 to 372800
Data columns (total 20 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   NTD ID                          372801 non-null  int64
 1   Legacy NTD ID                   366204 non-null  object
 2   Agency                          372801 non-null  str
 3   Mode/Type of Service Status     372801 non-null  str
 4   Reporter Type                   372801 non-null  str
 5   UACE CD                         372741 non-null  float64
 6   UZA Name                        372741 non-null  str
```

```
 7   Mode                        372801 non-null  str
 8   TOS                         372801 non-null  str
 9   3 Mode                      372801 non-null  str
 10  Date                        372801 non-null  datetime64[us]
 11  UPT                         372801 non-null  str
 12  VOMS                        371086 non-null  object
 13  VRH                         369288 non-null  str
 14  VRM                         369519 non-null  str
 15  Agency Mode TOS Date        372801 non-null  str
 16  State                       372801 non-null  str
 17  FTA Region                  372801 non-null  int64
 18  UPT_num                     372801 non-null  int64
 19  VRM_num                     369519 non-null  float64
dtypes: datetime64[us](1), float64(2), int64(3), object(2), str(12)
memory usage: 109.1+ MB
```

[18]:
```
#Filter main dataset for top 10 ridership agencies based on ridership and Heavy␣
↪Rail (HR) / Light Rail (LR) only, removing buses, ferries as the␣
↪concentration is on rail transit
```

[19]:
```
top_10_agencies = df.groupby('Agency')['UPT_num'].sum().
↪sort_values(ascending=False).head(10).index.tolist()
print("Top 10 Agencies Automatically Extracted:")
for i, ag in enumerate(top_10_agencies, 1):
    print(f"Rank {i}: {ag}")
```

```
Top 10 Agencies Automatically Extracted:
Rank 1: MTA New York City Transit
Rank 2: Chicago Transit Authority
Rank 3: Los Angeles County Metropolitan Transportation Authority
Rank 4: Washington Metropolitan Area Transit Authority
Rank 5: Massachusetts Bay Transportation Authority
Rank 6: Southeastern Pennsylvania Transportation Authority
Rank 7: New Jersey Transit Corporation
Rank 8: City and County of San Francisco
Rank 9: County of Miami-Dade
Rank 10: Metropolitan Atlanta Rapid Transit Authority
```

[20]:
```
rail_df = df[
    (df['Agency'].isin(top_10_agencies)) &
    (df['Mode'].isin(['HR', 'LR']))
].copy()
```

[21]:
```
rail_df = rail_df.dropna(subset=['UPT_num', 'VRM_num'])
```

[22]:
```

```

```
# When you combine all the active months across those specific Heavy Rail and␣
 ↪Light Rail modes for your top 10 agencies-and throw out any months with␣
 ↪missing reporting-you are left with 3,968 complete, analysis-ready monthly␣
 ↪data points.
```

[23]:
```
# Verification of removed null values
```

[24]:
```
rail_df.info()
```

```
<class 'pandas.DataFrame'>
Index: 3968 entries, 28999 to 352089
Data columns (total 20 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   NTD ID                        3968 non-null   int64
 1   Legacy NTD ID                 3968 non-null   object
 2   Agency                        3968 non-null   str
 3   Mode/Type of Service Status   3968 non-null   str
 4   Reporter Type                 3968 non-null   str
 5   UACE CD                       3968 non-null   float64
 6   UZA Name                      3968 non-null   str
 7   Mode                          3968 non-null   str
 8   TOS                           3968 non-null   str
 9   3 Mode                        3968 non-null   str
 10  Date                          3968 non-null   datetime64[us]
 11  UPT                           3968 non-null   str
 12  VOMS                          3968 non-null   object
 13  VRH                           3968 non-null   str
 14  VRM                           3968 non-null   str
 15  Agency Mode TOS Date          3968 non-null   str
 16  State                         3968 non-null   str
 17  FTA Region                    3968 non-null   int64
 18  UPT_num                       3968 non-null   int64
 19  VRM_num                       3968 non-null   float64
dtypes: datetime64[us](1), float64(2), int64(3), object(2), str(12)
memory usage: 1.3+ MB
```

[25]:
```
rail_df.head()
```

[25]:

	NTD ID	Legacy NTD ID	Agency	Mode/Type of Service Status	Reporter Type	UACE CD	UZA Name
28999	10003	1003	Massachusetts Bay Transportation Authority	Active	Full Reporter	9271.0	Boston, MA--NH
29000	10003	1003	Massachusetts Bay Transportation Authority				
29001	10003	1003	Massachusetts Bay Transportation Authority				
29002	10003	1003	Massachusetts Bay Transportation Authority				
29003	10003	1003	Massachusetts Bay Transportation Authority				

```
29000                    Active  Full Reporter   9271.0  Boston, MA--NH
29001                    Active  Full Reporter   9271.0  Boston, MA--NH
29002                    Active  Full Reporter   9271.0  Boston, MA--NH
29003                    Active  Full Reporter   9271.0  Boston, MA--NH

      Mode TOS 3 Mode       Date         UPT VOMS     VRH        VRM  \
28999   HR  DO   Rail 2002-01-01   9,952,800  320  75,194  1,654,277
29000   HR  DO   Rail 2002-02-01   9,753,800  320  68,228  1,501,014
29001   HR  DO   Rail 2002-03-01  10,469,800  320  75,194  1,654,277
29002   HR  DO   Rail 2002-04-01  10,736,100  320  74,761  1,644,740
29003   HR  DO   Rail 2002-05-01  10,526,600  320  76,110  1,674,419

                                    Agency Mode TOS Date State  FTA Region  \
28999  10003_HR_DO_Active_Full Reporter_09271_2002-01-01    MA           1
29000  10003_HR_DO_Active_Full Reporter_09271_2002-02-01    MA           1
29001  10003_HR_DO_Active_Full Reporter_09271_2002-03-01    MA           1
29002  10003_HR_DO_Active_Full Reporter_09271_2002-04-01    MA           1
29003  10003_HR_DO_Active_Full Reporter_09271_2002-05-01    MA           1

        UPT_num    VRM_num
28999   9952800  1654277.0
29000   9753800  1501014.0
29001  10469800  1654277.0
29002  10736100  1644740.0
29003  10526600  1674419.0
```

[26]:
```
rail_df.shape
```

[26]:
```
(3968, 20)
```

[27]:
```
print(rail_df.groupby(['Agency', 'Mode']).size())
```

```
Agency                                                  Mode
Chicago Transit Authority                               HR      296
City and County of San Francisco                        LR      296
County of Miami-Dade                                    HR      296
Los Angeles County Metropolitan Transportation Authority HR     296
                                                        LR      296
MTA New York City Transit                               HR      296
Massachusetts Bay Transportation Authority              HR      296
                                                        LR      296
Metropolitan Atlanta Rapid Transit Authority            HR      296
New Jersey Transit Corporation                          LR      592
Southeastern Pennsylvania Transportation Authority      HR      296
                                                        LR      120
Washington Metropolitan Area Transit Authority          HR      296
dtype: int64
```

Possible issues in Time-Series Length Across Agencies

When examining the record counts per agency and mode (e.g., New Jersey Transit showing ~592 monthly records for Light Rail versus ~296 for other agencies), two key structural factors explain these variances:

1. Multiple Reporting Sub-Segments Major regional authorities like New Jersey Transit operate multiple distinct light rail divisions or separate operating segments that report to the National Transit Database (NTD) as individual entries. When filtered, these overlapping sub-networks run concurrently, doubling the expected monthly row count.
2. Data Coverage Some agencies initiated federal reporting later or experienced reporting changes leading to shorter continuous blocks of monthly data (e.g., SEPTA Light Rail showing 120 months).

Impact

These differences do not invalidate the analysis, but they highlight the standardizing multi-modal state should adhere to.

[28]:
```
#Exploratory Data Analysis (EDA) & Summary Statistics
```

[29]:
```
summary_stats = rail_df.groupby(['Agency', 'Mode']).agg(
    Record_Count=('UPT_num', 'count'),
    Mean_UPT=('UPT_num', 'mean'),
    Median_UPT=('UPT_num', 'median'),
    Std_UPT=('UPT_num', 'std'),
    Min_UPT=('UPT_num', 'min'),
    Max_UPT=('UPT_num', 'max'),
    Total_UPT=('UPT_num', 'sum'),
    Total_VRM=('VRM_num', 'sum')
).reset_index()
```

[30]:
```
summary_stats['Trips_per_VRM'] = summary_stats['Total_UPT'] /␣
↪summary_stats['Total_VRM']
```

[31]:
```
# Order the output by the total ridership (UPT) with the highest agency, NYC␣
↪MTA at the top
```

[32]:
```
summary_stats.sort_values(by='Total_UPT', ascending=False)
```

[32]:

	Agency	Mode	Record_Count	\
5	MTA New York City Transit	HR	296	
12	Washington Metropolitan Area Transit Authority	HR	296	
0	Chicago Transit Authority	HR	296	
6	Massachusetts Bay Transportation Authority	HR	296	
10	Southeastern Pennsylvania Transportation Autho…	HR	296	
8	Metropolitan Atlanta Rapid Transit Authority	HR	296	
7	Massachusetts Bay Transportation Authority	LR	296	
4	Los Angeles County Metropolitan Transportation…	LR	296	

```
1                    City and County of San Francisco   LR           296
3   Los Angeles County Metropolitan Transportation…   HR           296
9                     New Jersey Transit Corporation   LR           592
2                               County of Miami-Dade    HR           296
11  Southeastern Pennsylvania Transportation Autho…   LR           120

        Mean_UPT   Median_UPT       Std_UPT   Min_UPT    Max_UPT    Total_UPT  \
5   1.912245e+08  197531202.0  3.868086e+07  20254269  253609943  56602441362
12  1.882599e+07   20458319.0  6.174360e+06   1182655   27860355   5572493503
0   1.531147e+07   16298314.5  4.477810e+06   2235078   22663990   4532195679
6   1.125566e+07   12136002.0  3.183021e+06   1182140   16679922   3331676310
10  6.893006e+06    7301716.0  1.775097e+06   1263591    9821204   2040329682
8   5.078523e+06    5689745.5  1.722154e+06    993578    7982627   1503242849
7   4.971451e+06    5333057.5  1.571460e+06    259090    7543900   1471549430
4   3.942146e+06    3863438.0  1.099138e+06    725405    5958605   1166875287
1   3.416400e+06    3791224.0  1.120721e+06         0    5419319   1011254370
3   3.149356e+06    3509658.5  8.942861e+05    616954    4629231    932209447
9   7.431610e+05     504067.5  4.043196e+05     35811    1615969    439951286
2   1.403744e+06    1432737.5  3.035689e+05    287803    1996221    415508237
11  2.232625e+06    2267137.0  5.316367e+05   1177417    3438899    267915057

       Total_VRM  Trips_per_VRM
5   8.473828e+09       6.679678
12  1.802953e+09       3.090760
0   1.695204e+09       2.673540
6   5.379211e+08       6.193615
10  4.007416e+08       5.091385
8   5.162102e+08       2.912075
7   1.443906e+08      10.191446
4   3.186260e+08       3.662210
1   1.256476e+08       8.048335
3   1.530996e+08       6.088907
9   6.812246e+07       6.458241
2   1.872913e+08       2.218513
11  3.478319e+07       7.702428
```

[33]:
```
#charts and vivalization
```

Chart 1: Historical Monthly Ridership Trends (Heavy Rail vs. Light Rail) Goal: Evaluate long-term ridership trends from 2002 to present across the top 10 U.S. transit agencies.

Key Takeaways & Insights:

Dominance of Subways: Heavy Rail (subways) consistently generates higher passenger volume than Light Rail across all years.

COVID-19 Shock: Both trends for light rail and heavy rail show a massive drop in early 2020 due to pandemic stay-at-home orders, followed by a gradual post-covid recovery curve.

[34]:
```
plt.figure(figsize=(12, 6))
sns.lineplot(
    data=rail_df,
    x='Date',
    y='UPT_num',
    hue='Mode',
    estimator='sum',
    errorbar=None,
    linewidth=2.5
)
plt.title('Monthly Ridership Trends: Heavy Rail vs. Light Rail (Top 10␣
 ↪Agencies)')
plt.xlabel('Year')
plt.ylabel('Total Unlinked Passenger Trips (UPT)')
plt.legend(title='Rail Mode', labels=['Heavy Rail (Subway)', 'Light Rail'])
plt.show()
```

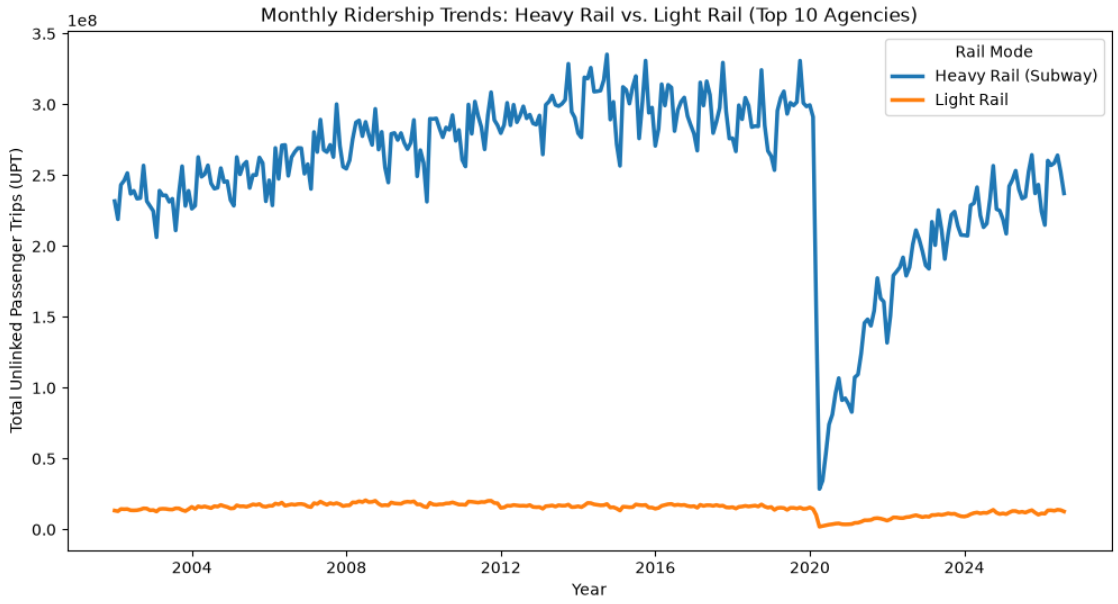


Chart 2: Vehicle Revenue Miles (VRM) vs. Ridership (UPT) Goal: Test the direct relationship between Vehicle Revenue Miles and Ridership.

Key Takeaways & Insights:

Strong Positive Linear Relationship: More service mileage correlates with more passenger trips, as seen in the linear cluster trending upward.

Modal Divergence: Heavy Rail data points cluster in the upper-right quadrant (high miles, extremely high ridership), while Light Rail remains concentrated in the lower-left quadrant (lower capacity/demand).

[35]:
```
plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=rail_df,
    x='VRM_num',
    y='UPT_num',
    hue='Mode',
    alpha=0.6,
    palette='Set1',
    s=50
)
plt.title('Service Supply (VRM) vs. Passenger Demand (UPT)')
plt.xlabel('Vehicle Revenue Miles (VRM - Service Supplied)')
plt.ylabel('Unlinked Passenger Trips (UPT - Ridership Demand)')
plt.legend(title='Rail Mode')
plt.show()
```

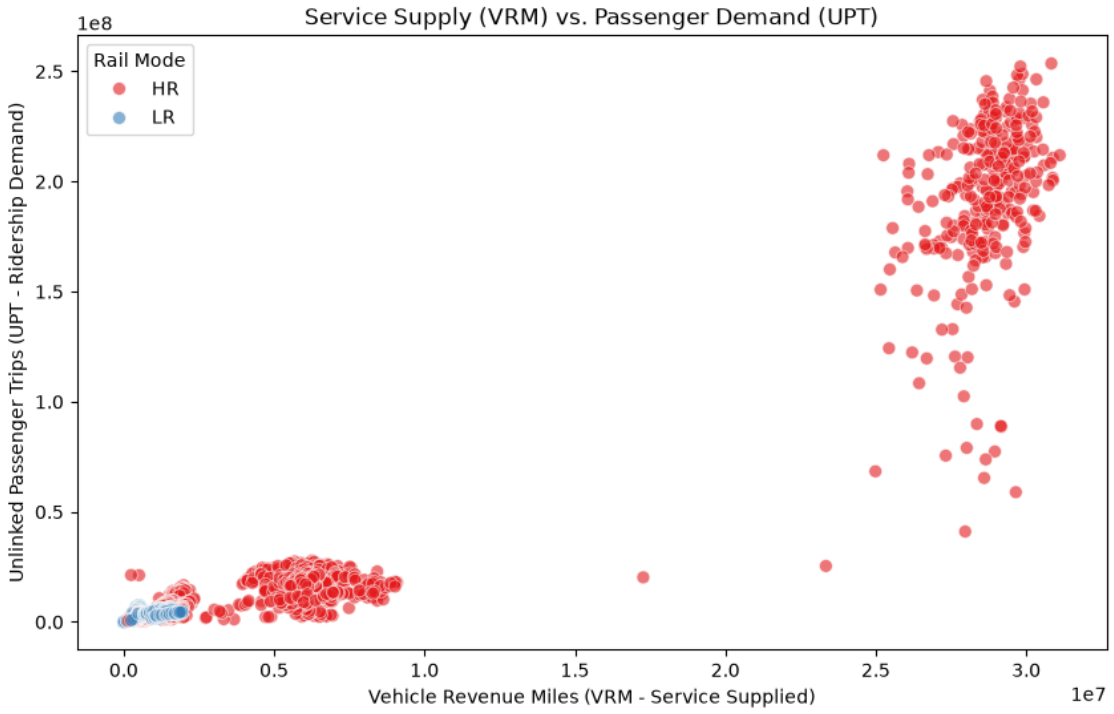


Chart 3: Operational Efficiency (Trips per Vehicle Revenue Mile)

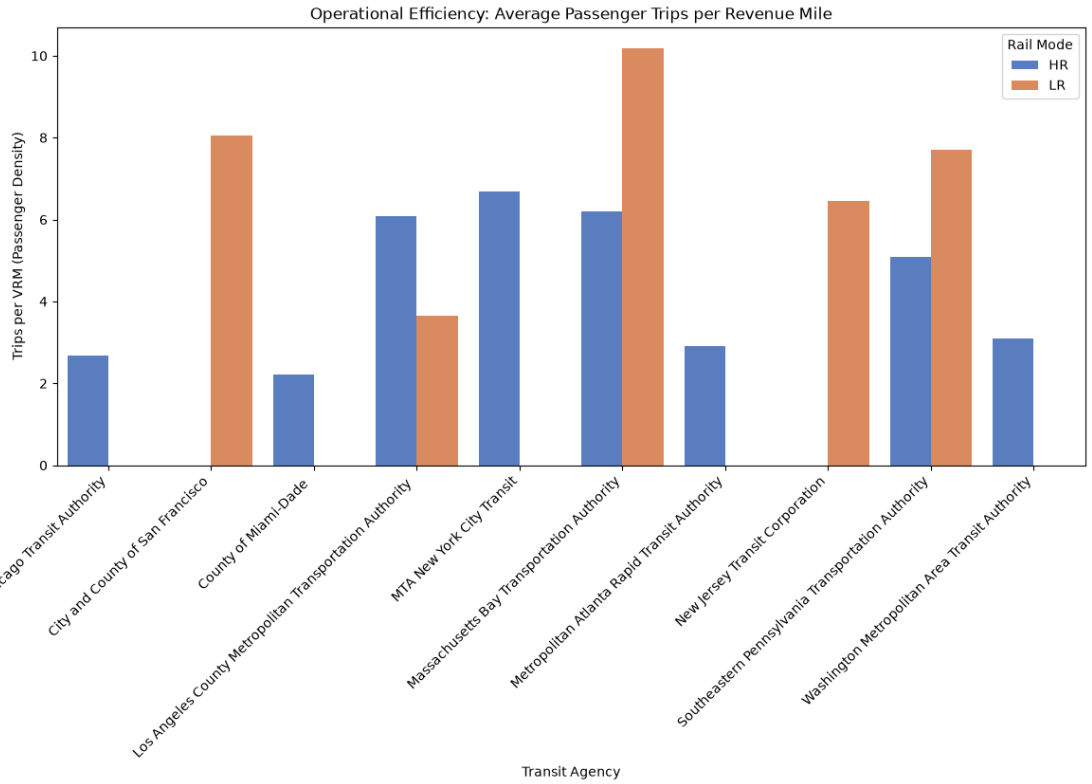
Goal: Compare passenger density across agencies to measure transit efficiency per service mile delivered.

Key Takeaways & Insights:Top Performer: MBTA (Boston) Light Rail achieves the highest efficiency ratio (~10.19 passenger trips per VR. Subway Efficiency: MTA New York City Transit demonstrates consistently strong subway efficiency (~6.68 trips per VRM)

[36]:
```
summary_chart = rail_df.groupby(['Agency', 'Mode']).agg(
    Total_UPT=('UPT_num', 'sum'),
    Total_VRM=('VRM_num', 'sum')
).reset_index()

summary_chart['Trips_per_VRM'] = summary_chart['Total_UPT'] /␣
 ↪summary_chart['Total_VRM']

plt.figure(figsize=(14, 6))
sns.barplot(
    data=summary_chart,
    x='Agency',
    y='Trips_per_VRM',
    hue='Mode',
    palette='muted'
)
plt.title('Operational Efficiency: Average Passenger Trips per Revenue Mile')
plt.xlabel('Transit Agency')
plt.ylabel('Trips per VRM (Passenger Density)')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Rail Mode')
plt.show()
```



[37]:
```
# Extra Credit - a defensible train/test-evaluated model

# How effectively service supply can predict passenger demand
```

[38]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = rail_df[['VRM_num']]
y = rail_df['UPT_num']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
 ↪random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(f"R-squared Score: {r2_score(y_test, y_pred):.4f}")
print(f"Coefficient (Slope): {model.coef_[0]:.4f}")
print(f"Intercept: {model.intercept_:.4f}")
```

```
R-squared Score: 0.9140
Coefficient (Slope): 6.6042
Intercept: -4063780.9596
```

Regression Model Results & Interpretation

R^2 Score of (0.9140) indicates a strong linear fit. 91.4% of passenger ridership is explained by vehicle revenue miles, proving that service supply is a primary driver of ridership demand.

Slope (6.60) suggests that for every single mile a train runs, the system get an average of about 6.6 passenger boardings added.

Intercept of (-4,063,781) shows the model's mathematical baseline. The negative intercept is a mathematical artifact of linear extrapolation. Because the dataset focuses on the top 10 largest transit agencies, the model lacks baseline data near zero. In reality, passenger demand floors out at zero rather than dropping into negative values, demonstrating why linear models should not be interpreted literally at extreme baseline values.

Ethics, Bias, and Limitations

Pandemic Distortion National transit data includes the massive drop in ridership during 2020–2021 and subsequent recovery patterns, which can heavily skew linear regression trends and long-term forecasts.

Data Reporting Inconsistencies Different agencies may define Ridership or Vehicle Revenue Miles differently based on counting technologies like fares, gate entry, and APC (automated passenger counters).

Equity & Service Supply Vehicle Revenue Miles does not always equal equitable access. Transit agencies for the rail has historically been used in richer neighborhoods in contrast to the bus lines

that cities run. So there can be a limitation in the report as this one only covers the rail system and not the bus fleet of each agency.

AI Prompt

1) All my charts are difficult to see and are jumbled together, what is a good way to display the data for easy reading? Implemented size recommendations so the chart is readable and included a legend for more clarity.

2) How can I seperate the lines for heavy rail and light rail to show that on the same chart without spreading it? Implemented line with 'hue' that was able to place the light rail lines into the chart.